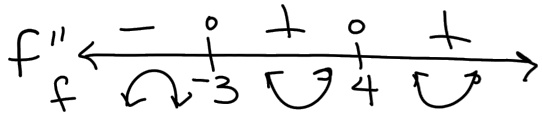
Worksheet 5-6 A Concavity and Points of Inflection #1
AP Calculus

Name SOLUTIONS

**Directions:** For each of the following, identify the intervals of concavity for $f(x)$ and find any value(s) of $x$ where $f(x)$ has a point of inflection. DO ALL WORK ON NOTEBOOK PAPER

1. $f''(x) = 5(x-4)^2(x+3)$




Concave up: $(-3,4) \cup (4,+\infty)$
because $f''(x) > 0$

Concave down: $(-\infty,-3)$
because $f''(x) < 0$

Pts of inflection at $x=-3$
because $f''(x)=0$ and $f''(x)$ changes signs

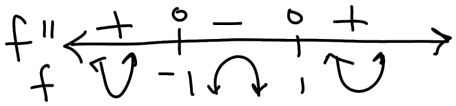
2. $f(x) = x^4 - 6x^2$

$f'(x) = 4x^3 - 12x$

$f''(x) = 12x^2 - 12 = 0$

$12(x^2-1) = 0$

$12(x+1)(x-1)$



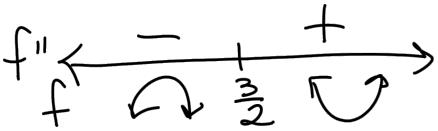

Concave up: $(-\infty,-1) \cup (1,+\infty)$
because $f''(x) > 0$

Concave down: $(-1,1)$
because $f''(x) < 0$

Pts of inflection at $x=-1$, $x=1$
because $f''(x)=0$ and $f''(x)$ changes signs

3. $f'(x) = x^2 - 3x - 10$

$f''(x) = 2x - 3$




Concave up: $(\frac{3}{2},+\infty)$
because $f''(x) > 0$

Concave down: $(-\infty,\frac{3}{2})$
because $f''(x) < 0$

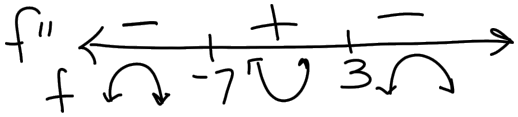
Pts. of inflection at $x=\frac{3}{2}$
because $f''(x)=0$ and $f''(x)$ changes signs

4. $f'(x) = -x^3 - 6x^2 + 63x - 2$

$f''(x) = -3x^2 - 12x + 63 = 0$

$-3(x^2+4x-21) = 0$

$-3(x+7)(x-3) = 0$



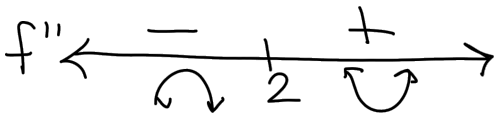

Concave up: $(-7,3)$
because $f''(x) > 0$

Concave down: $(-\infty,-7) \cup (3,+\infty)$
because $f''(x) < 0$

Pts of inflection at $x=-7$ and $x=3$
because $f''(x)=0$ and $f''(x)$ changes signs

6. $f'(x) = e^{x^2-4x}$

$$f''(x) = e^{(x^2-4x)}(2x-4)$$




Concave up: $(2, +\infty)$
because $f''(x) > 0$

Concave down: $(-\infty, 2)$
because $f''(x) < 0$

Pts of Inflection at $x = 2$
because $f''(x) = 0$ and
$f''(x)$ changes signs

8. $f(x) = \dfrac{x}{x-3}$

$$f'(x) = \frac{(x-3)1 - x(1)}{(x-3)^2}$$

$$= \frac{x-3-x}{(x-3)^2}$$

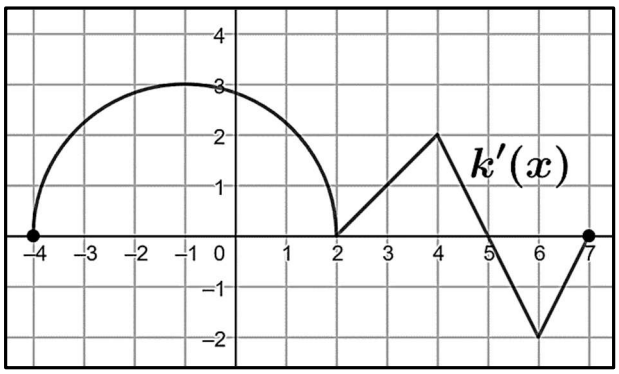
$$= \frac{-3}{(x-3)^2} = -3(x-3)^{-2}$$

$$f''(x) = 6(x-3)^{-3} = \frac{6}{(x-3)^3}$$

Concave up: $(3, +\infty)$
because $f''(x) > 0$

Concave down $(-\infty, 3)$
because $f''(x) < 0$

Pts of Inflection: none
$f(x)$ is undefined at $x = 3$




10. The function $k$ is continuous and differentiable on the interval $[-4, 7]$. The graph of $k'$, the derivative of $k$, consists of three line segments and a semicircle as shown in the figure above.

A) On what open interval(s) is $k(x)$ concave down? Give a reason for your answer.

$k(x)$ is concave down on $(-1, 2) \cup (4, 6)$ because $k'(x)$ is decreasing on these intervals.

B) For which value(s) of $x$ does $k(x)$ have a point of inflection? Give a reason for your answer.

$k(x)$ has a point of inflection at $x = -1$ and $x = 4$ because $k'(x)$ changes from increasing to decreasing, and at $x = 2$ and $x = 6$ because $k'(x)$ changes from decreasing to increasing at these $x$-values.

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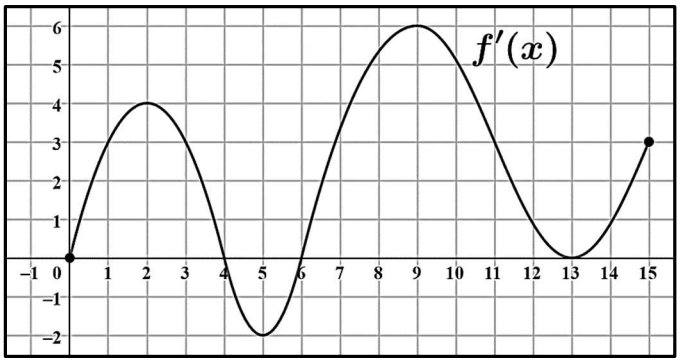


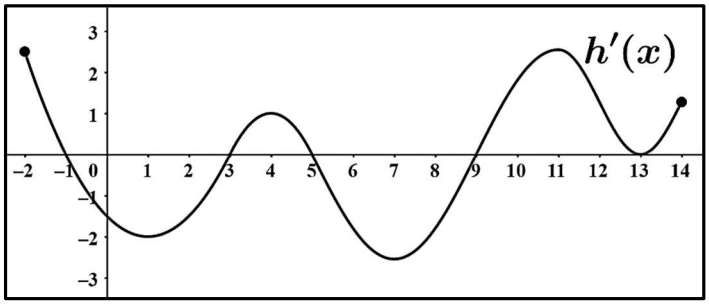

11. The function $f$ is continuous and twice differentiable on the interval $[0, 15]$. The graph of $f'$, the derivative of $f$, is given above and has horizontal tangents at $x = 2,\ 5,\ 9,$ and $13$.

A) On what interval(s) is $f(x)$ concave down? Explain your reasoning.

$f(x)$ is concave down on $(2, 5) \cup (9, 13)$ because $f'(x)$ is decreasing on these intervals.

B) For which value(s) of $x$ does $f(x)$ have a point of inflection? Give a reason for your answer.

$f(x)$ has a point of inflection at $x = 2$ and $x = 9$ because $f'(x)$ changes from increasing to decreasing, and at $x = 5$ and $x = 13$ because $f'(x)$ changes from decreasing to increasing.




12. The function $h$ is continuous and twice differentiable on the interval $[-2, 14]$. The graph of $h'$, the derivative of $h$, is given above and has horizontal tangents at $x = 1,\ 4,\ 7,\ 11,$ and 13.

A) For which value(s) of $x$ does $h(x)$ have a relative minimum? Give a reason for your answer.

$h(x)$ has a relative minimum at $x = 3$ and $x = 9$ because $h'(x)$ changes from negative to positive at these $x$-values.

B) For which value(s) of $x$ does $h(x)$ have a point of inflection? Give a reason for your answer.

$h(x)$ has a point of inflection at $x = 1$, $x = 7$, and $x = 13$ because $h'(x)$ changes from decreasing to increasing, and at $x = 4$, and $x = 11$ because $h'(x)$ changes from increasing to decreasing at these $x$-values.

C) On what interval(s) is the graph of $h(x)$ both decreasing and concave down? Explain your reasoning.

$h(x)$ is decreasing and concave down on $(-1, 1) \cup (5, 7)$ because $h'(x)$ is negative and decreasing on these intervals.